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AI-Powered Resume Analyzer

For Mechanical Engineering Candidates

Desktop Application · Python · Rule-Based AI

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Course	Artificial Intelligence
Course Code	BMEE407L
Assessment	Digital Assessment — 3a

GitHub Repository: github.com/vetri67/Resume-Analyzer-RoyalMechs

01 Introduction

This report documents the design, development, and evaluation of the Resume Analyzer for Royal Mechs — a desktop application built in Python that evaluates mechanical engineering resumes using rule-based AI logic. The tool provides automated scoring, structured feedback, and improvement suggestions through an intuitive GUI. This project fulfils the Digital Assessment 3 requirement for BMEE407L Artificial Intelligence at Vellore Institute of Technology, Chennai.

02 Objectives

- Develop a professional, GUI-based desktop application for resume analysis
- Implement rule-based AI logic to score key mechanical engineering competencies
- Provide structured, actionable feedback tailored for mechanical engineering students
- Demonstrate practical integration of AI concepts covered in BMEE407L
- Enable PDF resume upload with automated text extraction and keyword analysis

03 Key Features

Feature	Description
Modern GUI	Built with CustomTkinter — light-theme, tabbed card-style interface
PDF Resume Upload	File dialog + PyPDF2 to extract full resume text automatically
Tabbed Interface	Separate tabs for Skills input, Projects description, and Analysis output
AI Scoring Engine	Rule-based keyword matcher evaluating five mechanical engineering criteria
Animated Progress Bar	Threading-based score animation for engaging visual feedback
Emoji Feedback Panel	Colour-coded suggestions with specific, actionable improvement tips
Multi-source Input	Combines typed skills, project description, and extracted PDF text

04 Technologies Used

Library / Tool	Role
Python 3.x	Core programming language
CustomTkinter	Modern GUI framework for the desktop interface
PyPDF2	PDF text extraction from uploaded resumes
Pillow (PIL)	Image and logo rendering within the application
Threading	Non-blocking progress bar animation
Rule-Based Logic	AI scoring engine (keyword-based evaluation)
PyInstaller	Packaging the app as a standalone executable

05 Implementation

The application follows a clean single-file architecture, making it straightforward to deploy and share. The GUI is constructed with CustomTkinter, providing a modern tabbed interface. Uploaded PDF resumes are parsed by PyPDF2, lowercased, and combined with the user's typed input before being passed through the scoring engine, which evaluates five mechanical engineering criteria.

Scoring Criteria

Criterion	Keyword(s)	Points
Python Programming	python	15
CAD / Modelling Tools	SolidWorks, cad	15
Simulation — ANSYS	ANSYS	15
CFD Knowledge	CFD	15
Project Description Quality	> 20 characters	20
Maximum Total		80

** Score displayed out of 100; maximum achievable with current criteria is 80.*

Score Thresholds

Range	Rating	UI Colour
70 – 100	Excellent	Green
40 – 69	Good	Orange
0 – 39	Needs Improvement	Red

06 Application Screenshots

Figure 1 — Projects Tab (Home Interface)

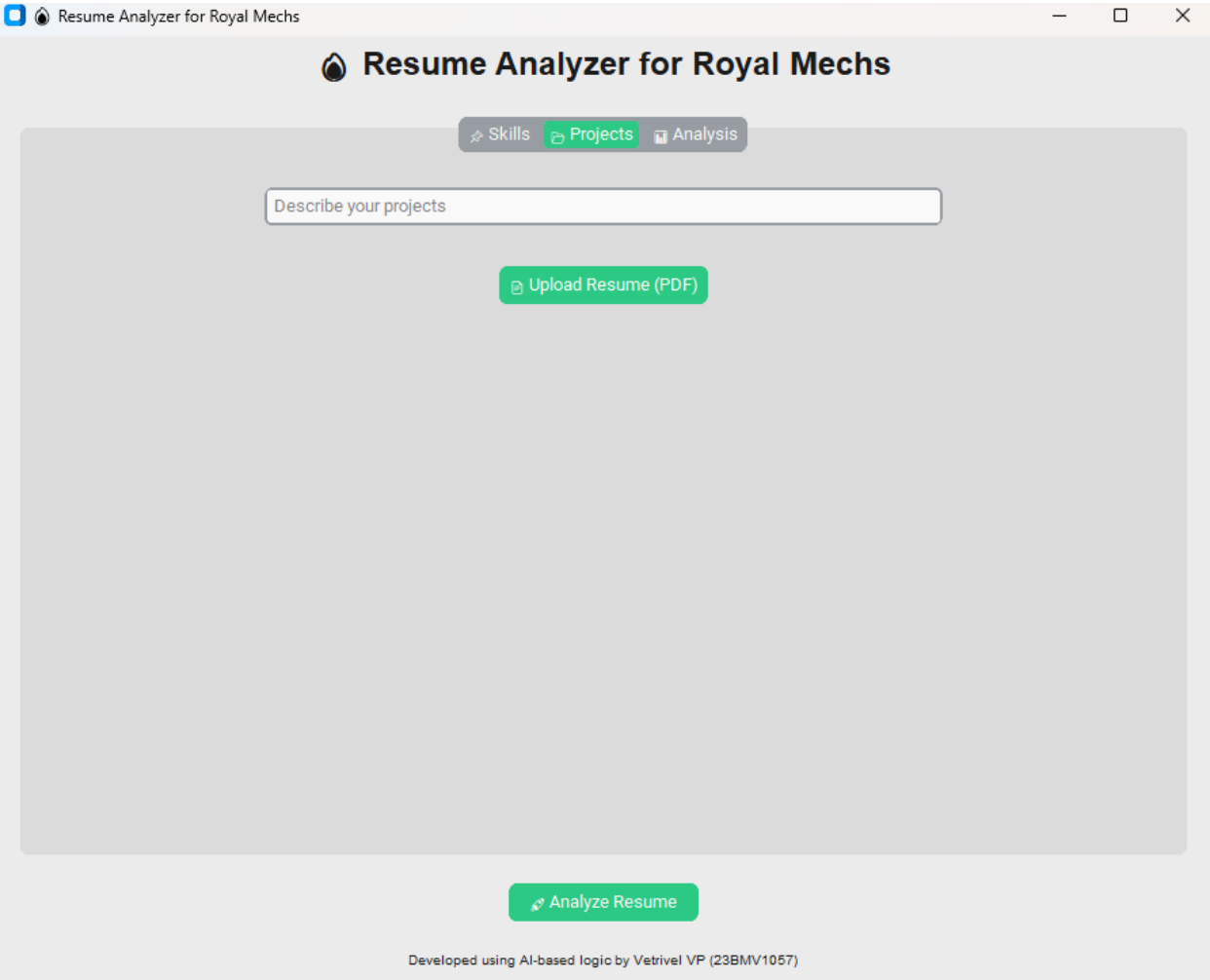
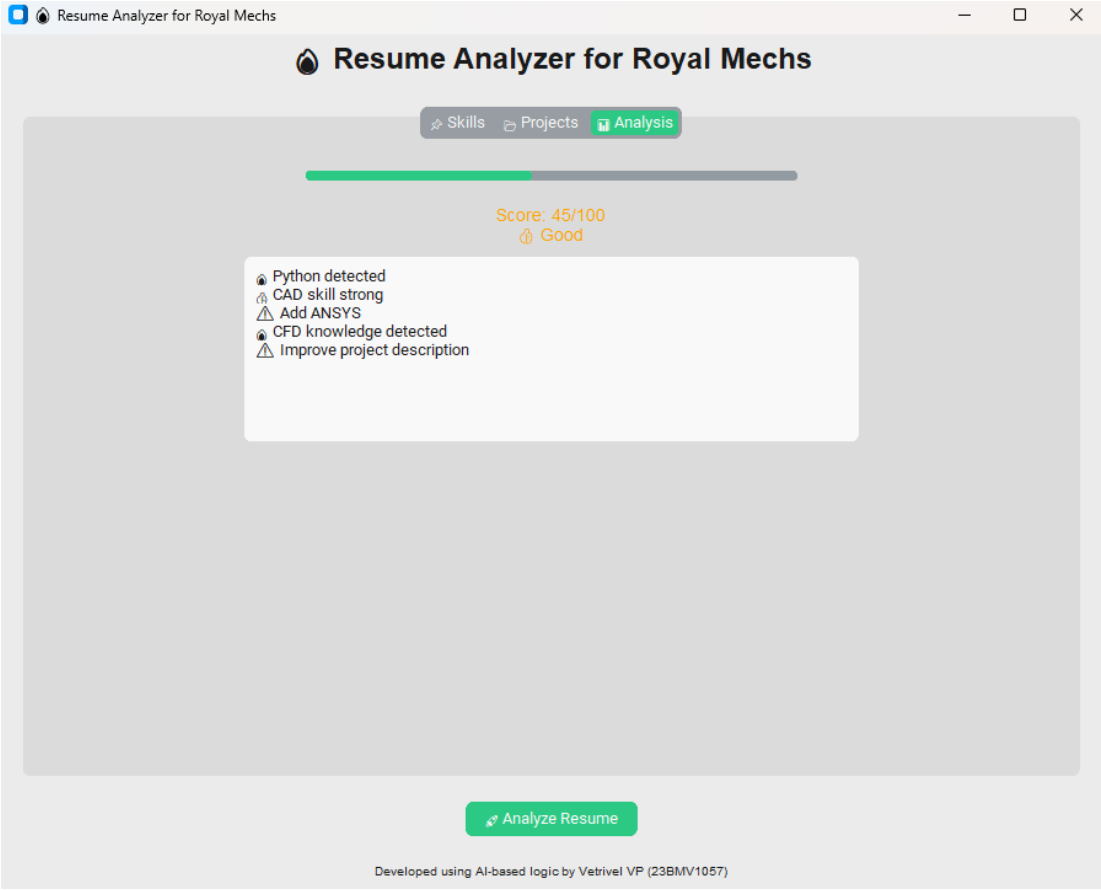


Figure 2 — Analysis Tab (Score & Feedback Output)



07 Results & Output

The application was tested with a sample mechanical engineering resume and returned a score of 45 / 100 (Good). Python and CAD skills were correctly detected; ANSYS and a sufficiently detailed project description were flagged for improvement. The animated progress bar filled to 45% and the feedback panel displayed colour-coded suggestions accordingly.

Check	Result	Points Awarded
Python detected	Pass	**+15**
CAD skill present	Pass	**+15**
ANSYS simulation	Not found	**+0**
CFD knowledge	Pass	**+15**
Project description	Too brief	**+0**
Total Score		**45 / 100**

08 Conclusion

The Resume Analyzer for Royal Mechs demonstrates how knowledge-based AI logic can be practically applied to an engineering problem. By combining PDF processing, keyword analysis, and an interactive GUI in a single Python application, it provides a functional and user-friendly resume evaluation tool. Future enhancements could include NLP-based semantic analysis, a broader skills database aligned with industry job descriptions, or web-based deployment for wider accessibility.

09 GitHub Respository

The complete source code, screenshots, and project files are available at:

Source Code: https://github.com/vetri67/Resume-Analyzer-RoyalMechs